COI- Assessment Tool - 2

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① Spectrum available = 45MHz
Each Simplex channel required 25KHZ.
Total Simplex channels = 45000 ÷ 25 = 1800.
two Simplex channels divide by 2 = 1800 ÷ 2 = 900.
Ans: 900 full - duplex channels.

② Total channels in the system = 560.
Frequency reuse cluster size = 7 cells.
Each cell allocation = $\frac{560}{7}$ = 80
Ans: 80 channels per cell.

③ GSM operator deploys 100 carriers.
Each speed = 200KHZ.
time = 8.
Total channels = 100 × 8 = 800.
Ans: 800 Simultaneous voice channels.

④ Power = 80W, handset power = 4W
Ratio = 80 ÷ 4 = 20.
dB = $10\log_{10}(20) \approx 13$.
Ans: 13dB Power diff.

⑤ Serving Signal length = -60dBm.
Co-channel interference = -92 dBm.
Carrier ratio = -60 - (-92) = 32
Ans: 32dB C/I.

Spectrum = 42MHz.

Channel band = 300 kHz.

Total channels = 42,000 ÷ 300 = 140.

140 ÷ 7 = 20.

Ans: 20 channels per cell.

T_x = 64W, Path-loss = 4.

Power = 2^4 = 16. New Power = 64 ÷ 16 = 4.

Ans: 4W

Pα $1/R^2$ = law.

radius = 2^2 = 4

Ans: 60 channels = 4x increases

Cluster size = 4, each cell = 15.

Total = 4x15 = 60

Ans: 60 channels

Speed = 100km/h ≈ 27.8m/s, Carrier = 900mHz.

Doppler = $\frac{v}{c} f \approx 83$.

Ans: ≈83Hz

TDMA frame = 5ms, slots = 8.

duration = 5 ÷ 8 = 0.625ms

Ans: 0.625ms

Spectrum = 24MHz, Spacing = 200 KHz

Carriers = 24,000 ÷ 200 = 120

Ans: 120 Carriers.

T_x = 8W = 39dBm, Pathloss = 33dB.

Received = 39-33 = 6

Ans: 6dBm.

(14) Distance = 8.5km, freq = 1900 MHz.

FSPL ≈ 92dB.

Ans: 98dBm

(15) cluster = 6 cells, each = 250 calls

Total = 6×250 = 1500.

Ans: 1500 calls.

(16) Call rate = 4000/s, avg duration = 180s.

Active calls = 4000 × 180 = 720,000

Ans: 720,000 calls.

(17) channels = 150, each = 25KHz.

Bandwith = 150×25 = 3750 KHz = 3.75 MHz.

Ans: 3.75 MHz.

(18) 30 bands → 29 guard gaps.

Total wasted = 29×40 = 1160 KHz = 1.16 MHz

Ans: 1.16 MHz

(19) Speed = 90 km/h = 1.5 km/min

Cell diameter = 3km → 2min/cell → 60/2 = 30

Ans: ~30 handoffs

(20) 8 centers × 40 stations × 250 users

Total = 80,000.

Ans: 80,000 users

(21) cell radius = 0.4 km, area ≈ 0.16 km²

Region = 400 ÷ 0.16 = 2500

Ans: ~2500 cells

(22) Spectrum = 6MHz, channel = 30kmHz.

channels = 6000 ÷ 30 = 200.

Ans: 200 channels.

(23) 1200 Subs x 4 calls/h = 4800 calls.
Avg length = 2.5 min → 12000 min/h ÷ 60 = 200 Erlangs.

Ans: 200 Erlangs.

(24) Bandwidth = 40 MHz, User = 400 KHz
Users = 40000 ÷ 400 = 100

Ans: 100 users

(25) 1200 stations x 600 users
Total = 720,000

Ans: 720,000 users

(26) 200 calls ÷ 4 per round = 50

Ans: 50 rounds.

(27) offered = 120,000, Capacity = 100,000
Blocked fraction = 20,000 ÷ 120000 = 16.7%.

Ans: ~16.7% blocked.

(28) 15% of 6,00,000 = 90,000.

Ans: 90,000 roaming.

(29) Load = 9000/s, Capacity = 12000/s.
Utillization = 9000 ÷ 12000 = 75%.

Ans: 75% utilized.

(30) Spectrum = 48 MHz ÷ 200 KHz = 240.
Reuse = 4 → 240 ÷ 4 = 60.

Ans: 60 channels/cell

(31) Spectrum = 30 MHz ÷ 200 KHz = 150.

Ans: 150 channels

(32) 2500 handoffs x 99% = 2475.

Ans: 2475 Successful.

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Ans:

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Ans:

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Ans:

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Ans:

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Ans:

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Ans:

11.

Ans:

12.

Ans:

(43) 150,000 ÷ 6000 = 25cycles.

Ans: 25cycles

(44) call length = 9min, hand off every 3min

Total = 3.

Ans = 3 hand offs

(45) Signal = -85dBm, noise = -115dBm.

SNR = 30dB.

Ans: 30dB.

(46) 250 x 150 Kbps = 37.5mbps.

Ans: 37.5 mbps

(47) 1500 subs x 4 calls/h = 6000 calls.

6000 x 2.5min = 15,000 min/h ÷ 60 = 250 Erlangs.

Ans: 250 Erlangs

(48) 250 x 12 KHz = 3MHz

Ans: 3MHz

(49) 1500 x 600 Kbps = 900 Mbps.

Ans: 900mbps

(50) Tower area = πR^2 = $\pi \times 16$ = 50.3 km^2

350 ÷ 50.3 = 7.

Ans: ~7 towers.